FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 16

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Recap *of previous lecture*

DIFFERENTIATION

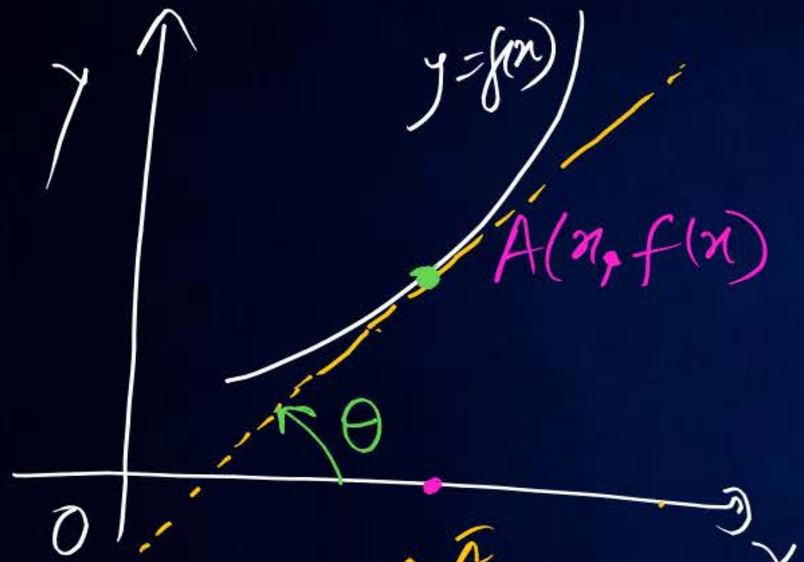


Topics *to be covered*

"MAXIMA & MINIMA"



$y = f(x)$ then $\frac{dy}{dx} = f'(x) = \tan\theta$ = Slope of tangent at any Random Point x on $f(x)$

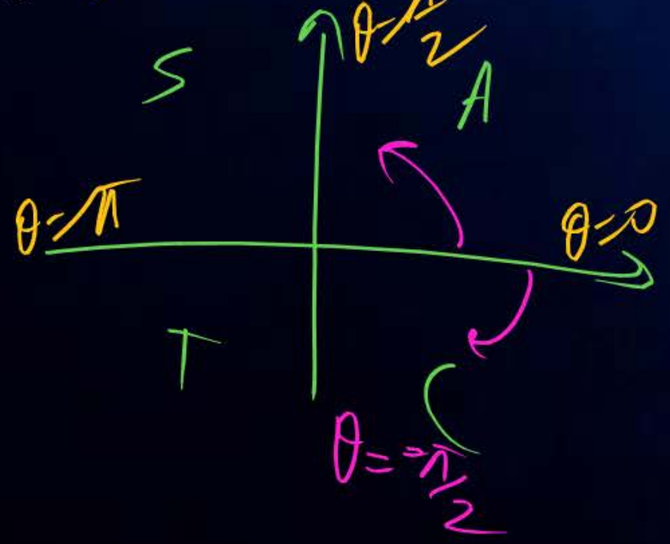


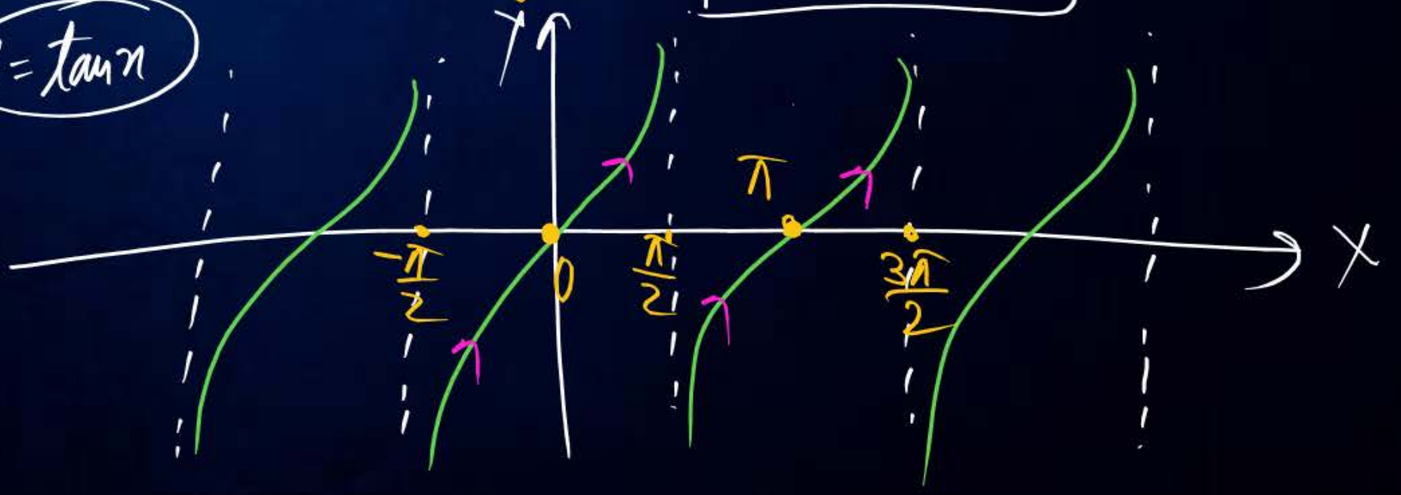
① for Acute angle $(0 < \theta < \pi/2)$ tangent, $f'(x) > 0$

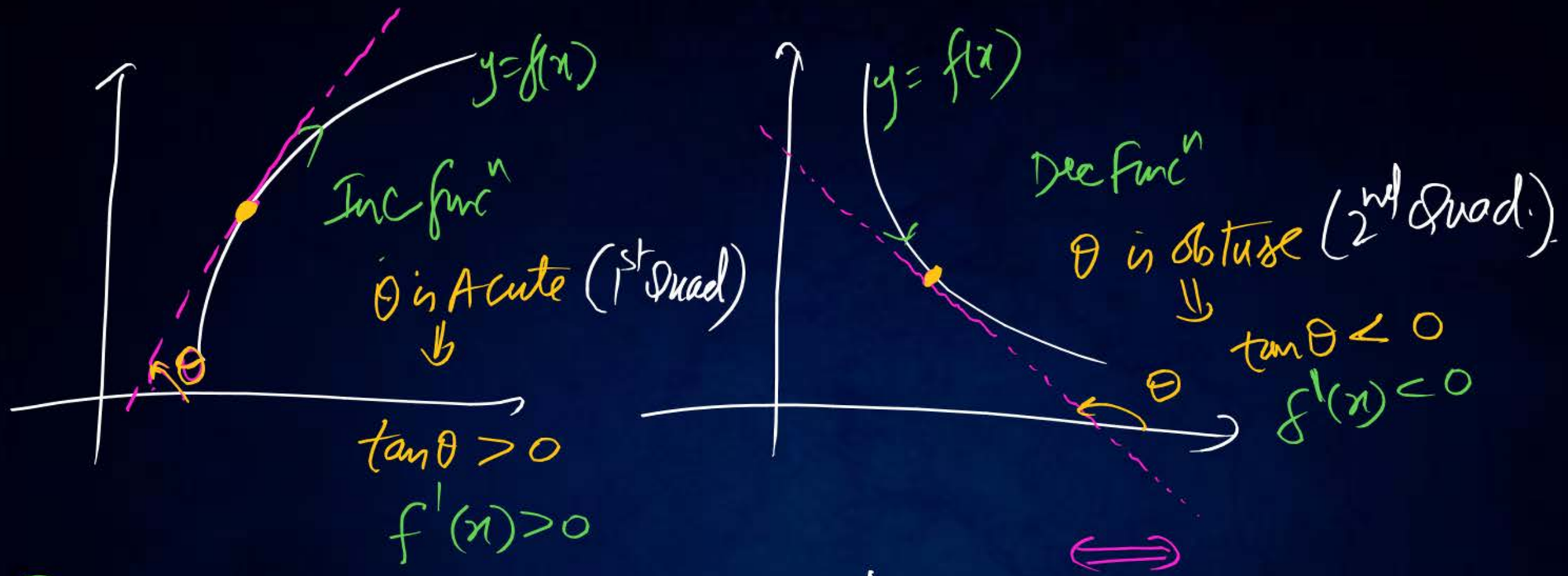
② for obtuse angle $(\frac{\pi}{2} < \theta < \pi)$ tangent, $f'(x) < 0$

③ For Horizontal tangent, $f'(x) = 0$

④ $y = \tan x$







⊛ for the func^n, $y=f(x)$, if $f'(x)>0$ then $f(x)$ is Increasing func^n.

& for the func^n, $y=f(x)$, if $f'(x)<0$ then $f(x)$ is Decreasing func^n.

Eg: Find the interval in which $f(x) = 2x^3 - 9x^2 + 12x + 15$ Increases and Decreases

Sol: $f'(x) = 6x^2 - 18x + 12$

$f'(x) = 6(x^2 - 3x + 2)$

$\boxed{f'(x) = 6(x-1)(x-2)}$

Turning Points of $f(x)$ can be Calculated By Putting $f'(x) = 0$

$6(x-1)(x-2) = 0$

i.e $x = 1$ & 2 are T. Points

Case I: when $x < 1 \Rightarrow x < 2$ also

$\Rightarrow (x-1) < 0$ & $(x-2) < 0$

i.e $(x-1)(x-2) > 0 \Rightarrow f'(x) > 0$

i.e $f(x)$ is Increasing when $x < 1$

Case II: when $1 < x < 2$

$\Rightarrow (x-1) > 0$ & $(x-2) < 0$

or $(x-1)(x-2) < 0 \Rightarrow f'(x) < 0$

i.e $f(x)$ is Decreasing when $1 < x < 2$

Case III: when $x > 2$ $\Rightarrow x > 1$ also.

$\Rightarrow (x-2) > 0$ & $(x-1) > 0$

i.e. $(x-1)(x-2) > 0$ or $f'(x) > 0$

i.e. $f(x)$ is Increasing when $x > 2$

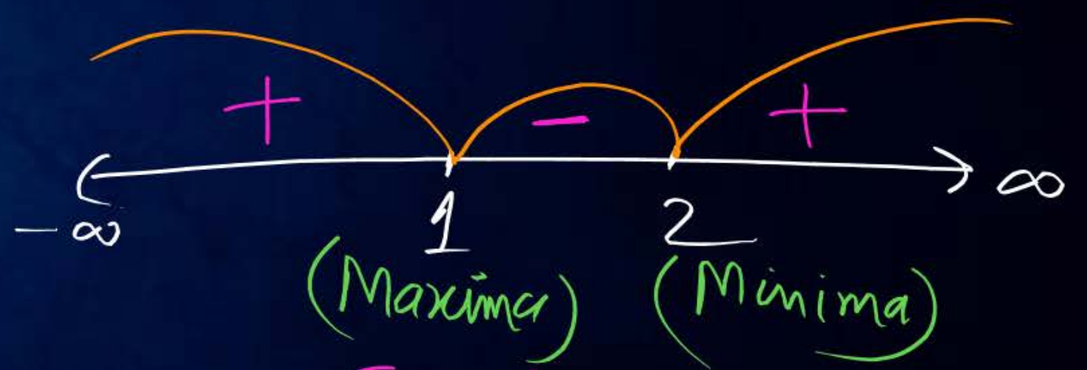
$\therefore f(x)$ is Inc in $(-\infty, 1) \cup (2, \infty)$

& $f(x)$ is Dec in $(1, 2)$

(M-II) $f(x) = 2x^3 - 9x^2 + 12x + 15$

$f'(x) = 6(x-1)(x-2)$

T. Points are: $x = 1$ & 2



Sign of $f'(x)$

$\therefore f(x)$ is Inc in $(-\infty, 1) \cup (2, \infty)$

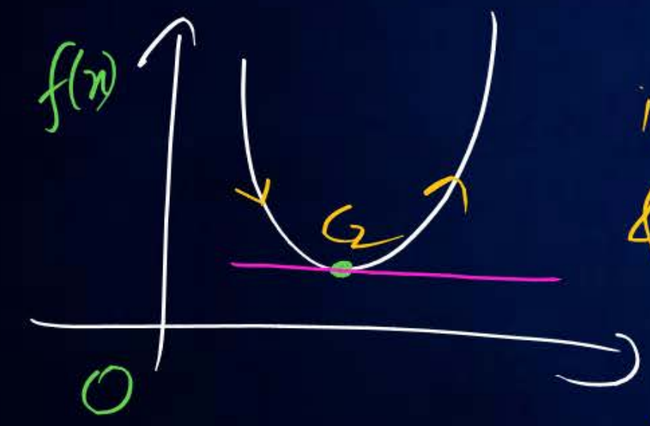
& $f(x)$ is Dec in $(1, 2)$

Maxima - Minima



N. Condⁿ for Maxima-Minima is $f'(x)=0 \Rightarrow x=C_1 \& C_2$

ie in the Left Nbd of C_1, $f(x)$ is Inc $\Rightarrow f'(x)>0$
& " Right " " , $f(x)$ is Dec $\Rightarrow f'(x)<0$ } Maxima



ie in the Left Nbd of C_2, $f(x)$ is Dec $\Rightarrow f'(x)<0$
& " Right " " , $f(x)$ is Inc $\Rightarrow f'(x)>0$ } Minima

Shortcut: let $x=C_1$ & C_2 are the T. Points then

we will check the Nature of $f'(x)$



C_1 (Maxima)



C_2 (Minima)

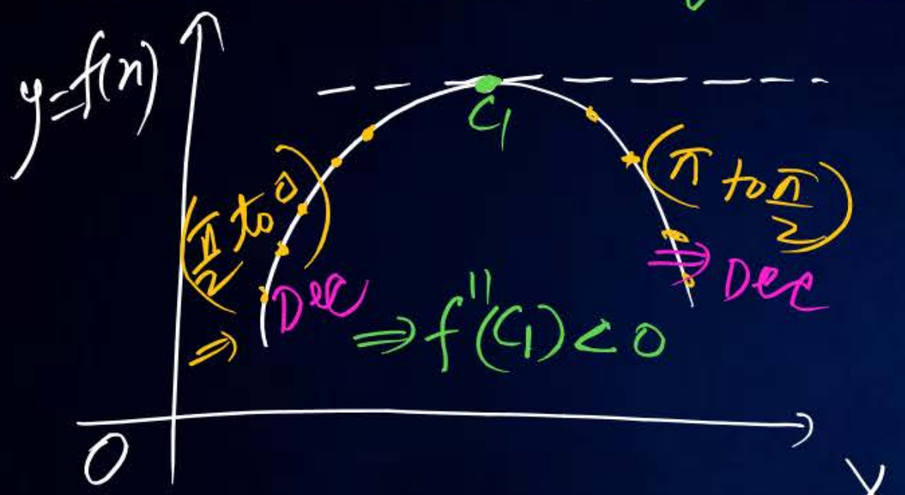


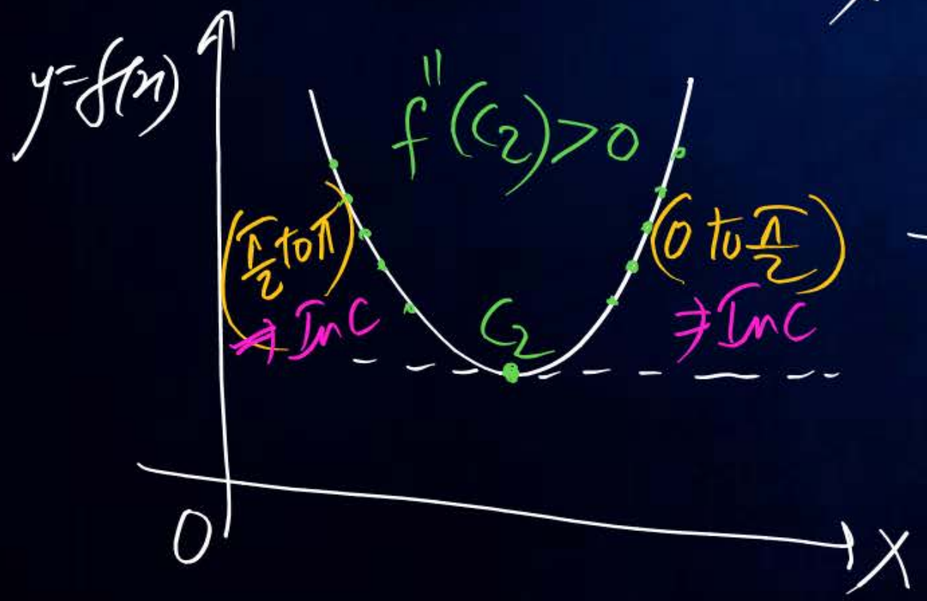
C_3 (NMNM)



C_4 (NMNM)

2nd Derivative Concept of Maxima-Minima



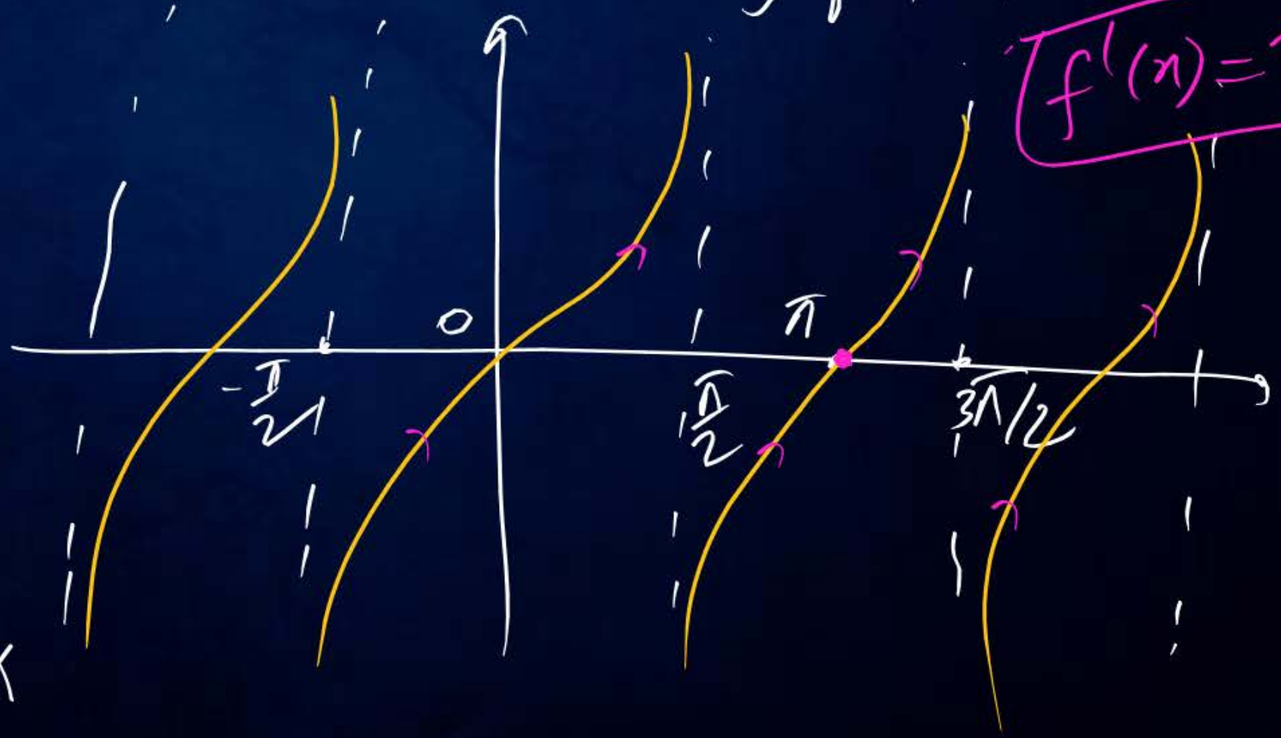


⊛ $f(x)$ is Inc if $f'(x) > 0$

$\Rightarrow f'(x)$ is Inc if $f''(x) > 0$

⊛ $f(x)$ is Decreasing if $f'(x) < 0$

$\Rightarrow f'(x)$ is Decreasing if $f''(x) < 0$



Hence at Critical Point c_1 if $f''(c_1) < 0$ then $x = c_1$ is the point of Maxima

& at " " c_2 if $f''(c_2) > 0$ " $x = c_2$ " " of Minima.

& Max. Value $= f(c_1) = ?$ & Minimum Value $= f(c_2) = ?$

eg $f(x) = 2x^3 - 9x^2 + 12x + 15$ then find Local Maxima & Local Minima of $f(x)$

Sol: $f'(x) = 6(x-1)(x-2) = 6(x^2 - 3x + 2)$

$\Rightarrow$ F Points are $x = 1$ & 2

$f''(x) = 6(2x - 3)$

Domain $= (-\infty, \infty)$

At $x = 1$, $f''(1) = -6$ ie -ve

So $x = 1$ is point of Maxima

&

at $x = 2$, $f''(2) = +6$ ie +ve

So $x = 2$ is point of Minima

Q: $f(x) = \dfrac{x}{\log_e x}$, Dom $= (0,\infty)$ except $x=1$

or $x>0$; $x\neq 1$ or $(0,1)\cup(1,\infty)$

$$f'(x) = \frac{\log x(1) - x\left(\frac{1}{x}\right)}{(\log_e x)^2} = \frac{\log_e x - 1}{(\log x)^2}$$

Critical points are, Put $f'(x)=0$

$$\frac{\log_e x - 1}{(\log_e x)^2} = 0$$

$$\log_e x - 1 = 0 \Rightarrow \log_e x = 1 \Rightarrow x = e^1 = e$$

i.e $x=e$ is the only C. Point.

sign of $f'(x) = \dfrac{\log_e x - 1}{(\log_e x)^2}$



(Minima)

$f(x)$ Decreasing in $(0,1)\cup(1,e]$

$f(x)$ Increasing in $[e,\infty)$

& Min Value $= f(e) = \dfrac{e}{\log_e e} = e$

$$f'(x) = \frac{\log_e x - 1}{(\log_e x)^2}$$

$$f(e^2) = \frac{2-1}{+ve} = +ve$$

$$f(2) = \frac{0.693 - 1}{+ve} = -ve$$

$$f\left(\frac{1}{e}\right) = \frac{\log_e\left(\frac{1}{e}\right) - 1}{(+ve)} = \frac{-1.\log_e e - 1}{+ve} = -ve$$

$$f''(x) = \frac{(\log_e x)^2 \cdot \left(\frac{1}{x}\right) - (\log_e x - 1)\left[2(\log_e x)\cdot\frac{1}{x}\right]}{\left[(\log_e x)^2\right]^2}$$

$$f''(e) = \frac{1 \times \frac{1}{e} - (1-1)\left(2 \times \frac{1}{e}\right)}{(1^2)^2} = \frac{1}{e} = +ve \Rightarrow \text{so } x = e \text{ is Point of Minima}$$

Q. $f(x) = \sin x + \cos x$ in $(0, \frac{\pi}{2})$ — (1)

Sol: $f'(x) = \cos x - \sin x$ — (2)

$f''(x) = -\sin x - \cos x$ — (3)

T. Points of $f(x)$ are Put $f'(x) = 0$

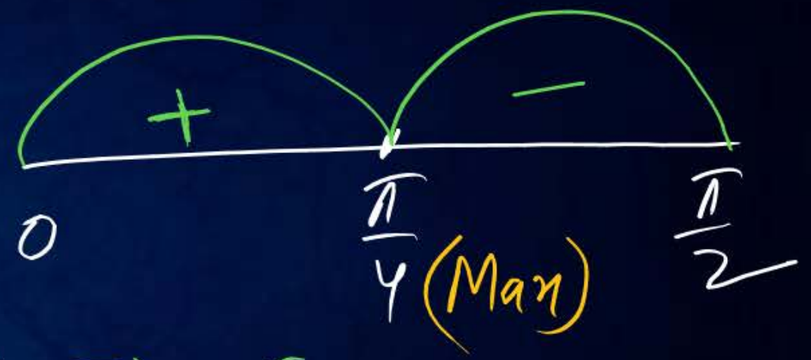
$\cos x - \sin x = 0$

$\tan x = 1 \Rightarrow x = \frac{\pi}{4}$

$f''(\frac{\pi}{4}) = -\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = -\sqrt{2}$

So $x = \frac{\pi}{4}$ is Point of Maxima (Max Value $= f(\frac{\pi}{4}) = \sqrt{2}$)

M-II $f'(x) = \cos x - \sin x$



$f'(\frac{\pi}{6}) = \frac{\sqrt{3}}{2} - \frac{1}{2} = +ve$

$f'(\frac{\pi}{3}) = \frac{1}{2} - \frac{\sqrt{3}}{2} = -ve$

So $f(x)$ Increases in $(0, \frac{\pi}{4}]$ & $f(x)$ Decreases in $[\frac{\pi}{4}, \frac{\pi}{2})$

Eg $f(x) = \sin 3x + 4$

We know that $-1 \le \sin 3x \le 1$

$-1+4 \le \sin 3x + 4 \le 1+4$

$3 \le f(x) \le 5$

i.e. $\text{Min } f(x) = 3$

& $\text{Max } f(x) = 5$

Eg $f(x) = -x^2 + 10$

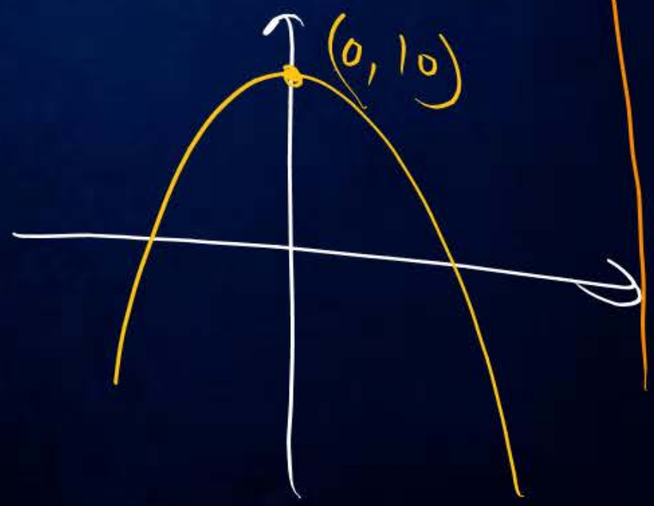
$\because x^2 \ge 0$

So $-x^2 \le 0$

$-x^2 + 10 \le 10$

$f(x) \le 10$

$-\infty < f(x) \le 10$

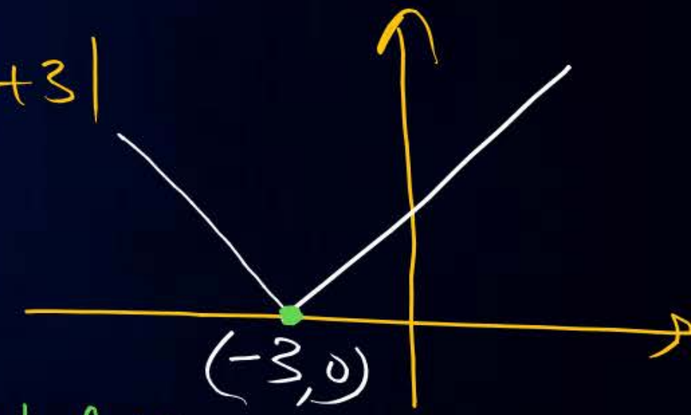


Eg $f(x) = |x+3|$

$y = |x|$



$y = |x+3|$



Point of Minima $= -3$

& Min Value $= 0$

Max Value $=$ DNE


Thank
YOU